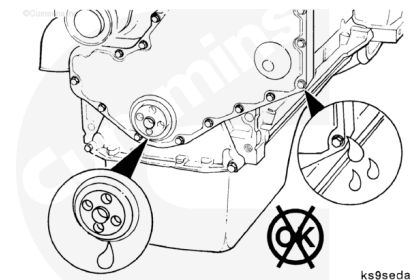


Trainings ▾

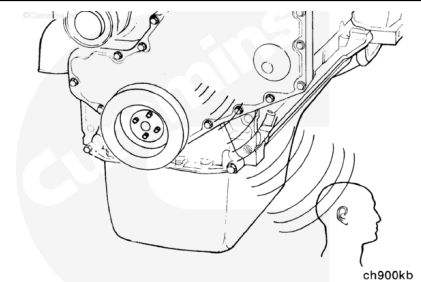
General Information

- Troubleshooting the front gear housing and gear train consists of checking for leaks at the gaskets (front cover, timing pin assembly, and fuel injection pump) and the front crankshaft lubricating oil seal, inspecting the gears, and measuring backlash, when required.
- Replace the front crankshaft oil seal. Refer to Procedure 001-023 (Crankshaft Seal, Front) in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-023.html>)



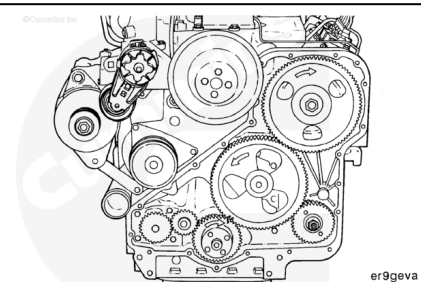
Gear noise emitted from the cover can indicate worn gear teeth.

Note : Excessive backlash can affect engine timing and engine performance.

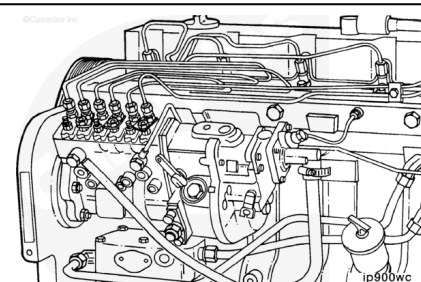


The gear train consists of:

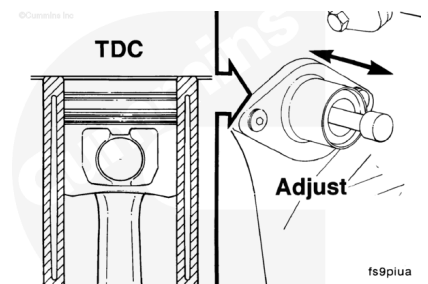
1. Crankshaft Gear
2. Lubricating Oil Pump Gear
3. Idler Gear
4. Camshaft Gear
5. Fuel Injection Pump Gear
6. Accessory Drive Gear; if used.



The gear housing provides a support for the fuel injection pump, the timing pin, and the accessory drive gear, if used.



Replacement of the gear housing or the timing pin assembly necessitates a realignment of the pin assembly on the housing to correspond to TDC for cylinder number 1. Refer to Procedure 001-049 (Timing Pin Housing) in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-049.html>)



Preparatory Steps

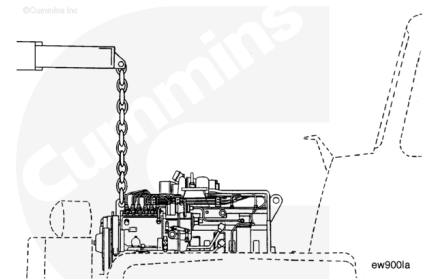
- Remove the drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-002.html>)
- Remove the vibration damper. Refer to Procedure 001-052 (Vibration Damper, Viscous) in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>)
- Remove the front gear cover. Refer to Procedure 001-031 (Accessory Drive Seal) in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-031.html>)
- Remove the accessory drive or air compressor, if equipped. Refer to Procedure 012-014 (Air Compressor) in Section 12. (</qs3/pubsys2/xml/en/procedures/100/100-012-014.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-011.html>)
- Remove the rocker lever assemblies. Refer to Procedure 003-008 (Rocker Lever) in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-008.html>)
- Remove the push rods. Refer to Procedure 004-014 (Push Rods or Tubes) in Section 4. (</qs3/pubsys2/xml/en/procedures/100/100-004-014.html>)
- Remove the camshaft. Refer to Procedure 001-008 (Engine Mounting Bolts) in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-008.html>)



- Remove the tappets. Refer to Procedure 004-015 (Tappet) in Section 4. (/qs3/pubsys2/xml/en/procedures/100/100-004-015.html)

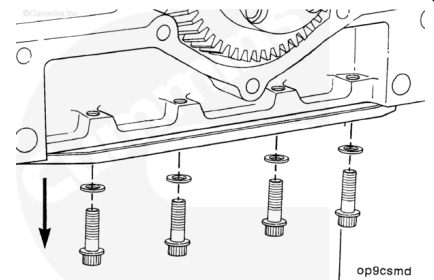
Remove

Support the front of the engine and remove the front engine mount.



⚠ CAUTION ⚠

Use extreme care when releasing the oil pan gasket from the gear housing to prevent damage to the gasket. If the gasket is damaged, it will probably be necessary to remove the oil pan and replace the gasket.

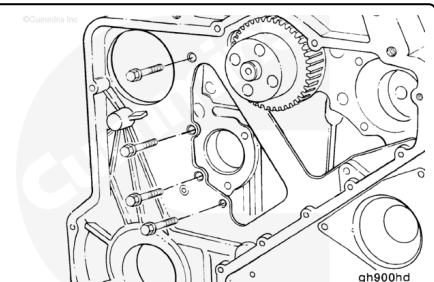


Loosen all the oil pan mounting capscrews four to five turns.

Remove the four front oil pan capscrews that secure the oil pan to the gear housing.

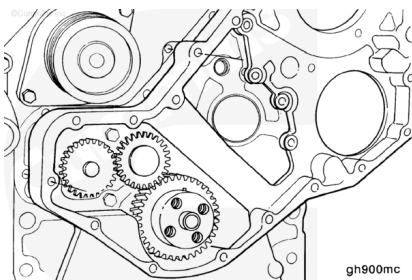
Insert a feeler gauge or a shim stock between the gear housing and the oil pan gasket. Move the feeler gauge or shim stock back and forth to release the gasket from the gear housing.

Remove the front gear housing capscrews.



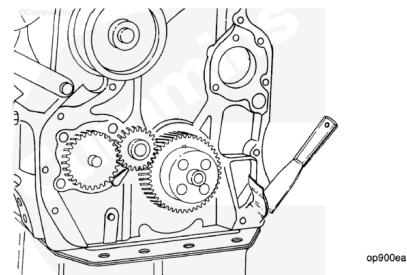
Remove the front gear housing.

-



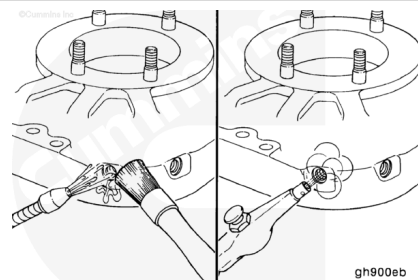
Clean

Clean the gasket material from the cylinder block.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

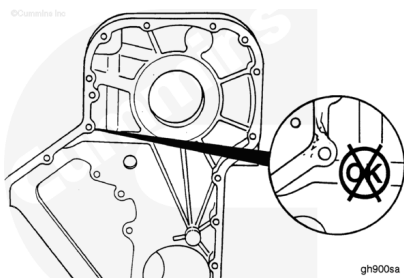
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Use solvent and a nylon bristle brush to clean the oil drillings.

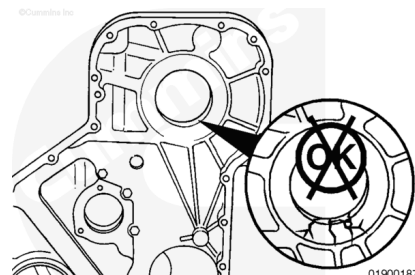
Dry the front gear housing with compressed air.

Inspect for Reuse

Inspect the front gear housing for cracks or damaged sealing surfaces.

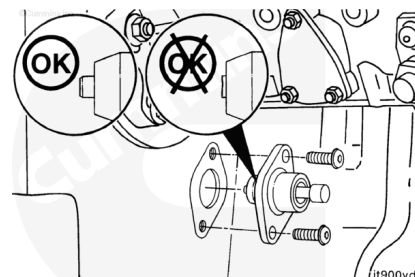


If the front gear housing is cracked at the accessory drive hole, evaluate the accessory/air compressor installation practices.



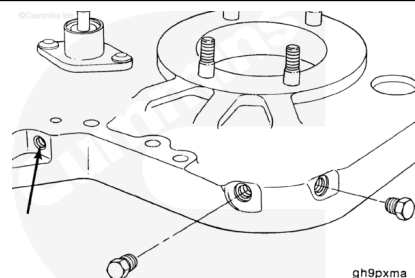
Inspect the timing pin housing and pin for damage.

Do **not** remove the timing pin assembly unless it is damaged or leaking, or if the gear housing is being replaced.

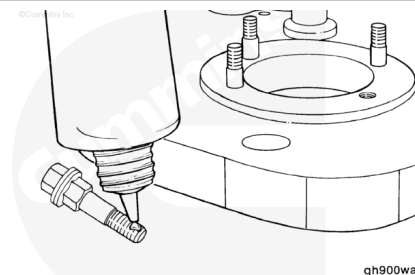


If the gear housing is being used again:

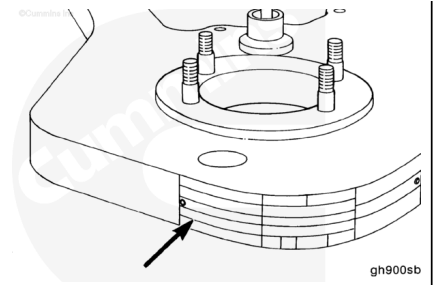
- Remove the pipe plugs from the oil drillings.



If the fuel injection pump studs are damaged or being installed in a new housing, coat the threads with Loctite® 242. Use two nuts, locked together, to remove and install the studs.



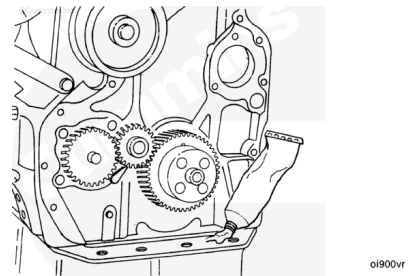
If the gear housing is being replaced, remove the engine data plate and install it on the new gear housing. Refer to Procedure 001-057 (Engine Data Plate) in Section 1.



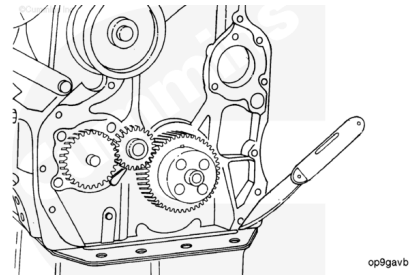
Install

The gear housing oil pan capscrews **must** be tightened within 15 minutes after applying the sealant.

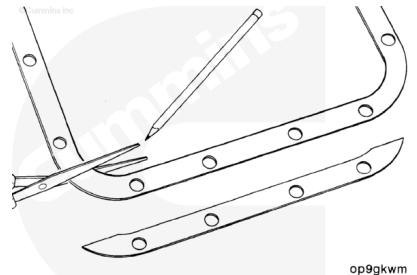
Inspect the oil pan gasket. If it is **not** torn, apply Three-Bond™ RTV sealant, Part Number 3823494, around the capscrew holes and at the joint of the gear housing.



If the pan gasket is torn, it can be repaired. Cut the torn gasket off even with the front of the cylinder block.

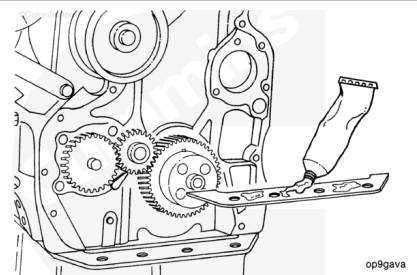


Use the old gasket as a pattern, cut the front section of a new gasket to the same size.

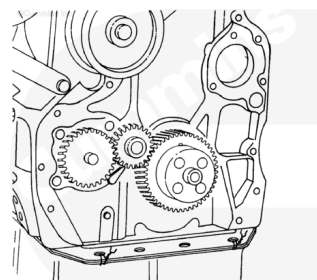


The gear housing and oil pan **must** be installed and tightened within 15 minutes after applying the sealant.

Clean the sealing surfaces and coat the new gasket on both sides with Three-Bond™ RTV sealant, Part Number 3823494.

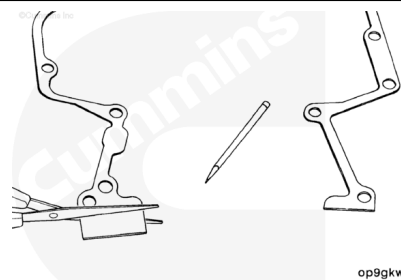


Use thread or a very fine wire to hold the new gasket splice in position as illustrated.



Mark and trim 1.6 mm [0.063 in] off the new gear housing gasket.

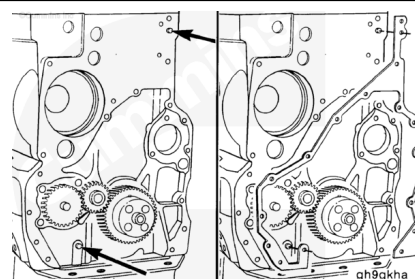
When properly trimmed, the gear housing gasket **must** be even with the lubricating oil pan gasket when installed.



Position the gasket on the alignment dowels.

Use guide pins (M8 x 1.25 x 50) to assist in aligning the gasket and gear housing.

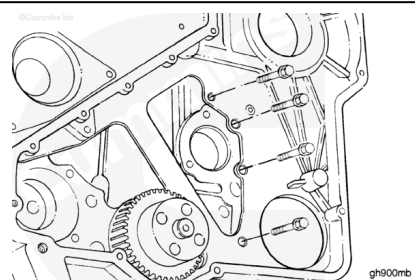
Make sure the guide pins are removed after alignment.



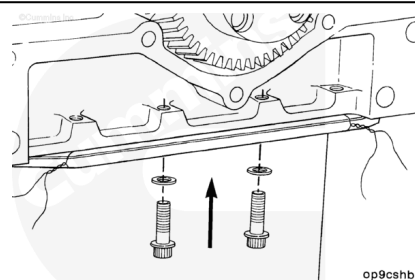
Carefully install the gear housing, making sure both gaskets are in place.

Install and tighten the gear housing mounting capscrews.

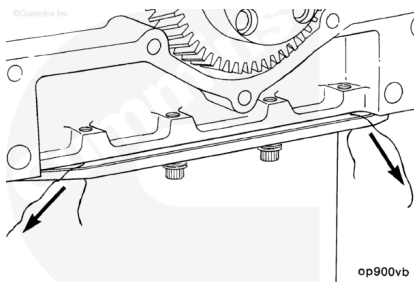
Torque Value: 24 n•m [212 in-lb]



Start the oil pan capscrews in the holes **not** being used to tie the gasket in place.



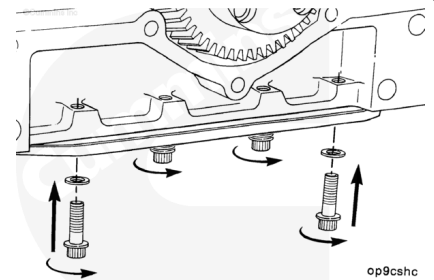
Remove the thread or wire holding the gasket in place.



Install the remaining two capscrews.

Tighten the oil pan cap screws.

Torque Value: 24 n•m [212 in-lb]



Finishing Steps

- Install the tappets. Refer to Procedure 004-015 (Tappet) in Section 4. (</qs3/pubsys2/xml/en/procedures/100/100-004-015.html>)
- Install the camshaft and camshaft thrust washer. Refer to Procedure 001-008 (Engine Mounting Bolts) in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-008.html>)
- Install the push rods. Refer to Procedure 004-014 (Push Rods or Tubes) in Section 4. (</qs3/pubsys2/xml/en/procedures/100/100-004-014.html>)
- Install the rocker levers. Refer to Procedure 003-008 (Rocker Lever) in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-008.html>) Adjust the valves. Refer to Procedure 003-004 (Overhead Set) in Section 3. (</qs3/pubsys2/xml/en/procedures/41/41-003-004.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-011.html>)
- Install the accessory drive or air compressor. Refer to Procedure 012-014 (Air Compressor) in Section 12. (</qs3/pubsys2/xml/en/procedures/100/100-012-014.html>).



- Install the front gear cover. Refer to Procedure 001-031 (Accessory Drive Seal) in Section 1.
(/qs3/pubsys2/xml/en/procedures/100/100-001-031.html)Install the vibration damper. Refer to Procedure 001-052 (Vibration Damper, Viscous) in Section 1.
(/qs3/pubsys2/xml/en/procedures/100/100-001-052.html)
- Install the drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-002.html).
- Operate the engine at idle for 5 to 10 minutes and check for leaks.